2.4 Mathematical Induction

Mathematical induction is one of the techniques which can be used to prove variety of mathematical statements which are formulated in terms of n, where n is a positive integer.

Mathematical Induction is used in the branches of Algebra, Geometry and Analysis where it turns out to be necessary to prove some truths of propositions.

The principle of mathematical induction:

Let $P(n)$ be a given statement for $n \in N$.

(i) Initial step: Let the statement is true for $n = 1$ i.e., $P(1)$ is true and

(ii) Inductive step: If the statement is true for $n = k$ (where k is a particular but arbitrary natural number) then the statement is true for $n = k + 1$. i.e., truth of $P(k)$ implies the truth of $P(k+1)$. Then $P(n)$ is true for all natural numbers n.

Example 2.34

Using mathematical induction method, Prove that

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2},\ n \in N.$$

Solution

Let the given statement $P(n)$ be defined as

$$1+2+3+\ldots+n = \frac{n(n+1)}{2} \text{ for } n \in N$$

Step 1: Put $n = 1$

LHS $P(1) = 1$

RHS $P(1) = \frac{1(1+1)}{2} = 1$

LHS = RHS for $n = 1$

$\therefore$ P(1) is true

Step 2: Let us assume that the statement is true for $n = k$.

i.e., P(k) is true

$$1 + 2 + 3 + \ldots + k = \frac{k(k+1)}{2} \text{ is true}$$

Step 3: To prove that $P(k + 1)$ is true

$$P(k+1) = 1 + 2 + 3 + \ldots + k + (k+1)$$

$$= P(k) + k + 1$$

$$= \frac{k(k+1)}{2} + k + 1$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$\therefore$ $P(k+1)$ is true

Thus if $P(k)$ is true, then $P(k+1)$ is also true.

$\therefore$ $P(n)$ is true for all $n \in N$

Hence $1+ 2 + 3 + \ldots + n = \frac{n(n+1)}{2}, n \in N$

Example 2.35

By the principle of Mathematical Induction, prove that

$1 + 3 + 5 + \ldots + (2n-1) = n^2$, for all $n \in N$.

Solution

Let $P(n)$ denote the statement

$1 + 3 + 5 + \ldots + (2n-1) = n^2$

Put $n = 1$

LHS = 1

RHS = 1^2

= 1

LHS = RHS

$\therefore$ $P(1)$ is true.

Assume that $P(k)$ is true.

i.e., $1 + 3 + 5 + \ldots + (2k-1) = k^2$

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